NLP Final Individual Report

Title: **Comparative Analysis: Threads and Twitter Reviews**

Completed

by: Snehitha

Tadapaneni

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Introduction

In this project, my teammates and I conducted a comprehensive analysis of Threads and Twitter app reviews using a multi-stage NLP pipeline consisting of data cleaning, sentiment modeling, and topic modeling. Our overall goal was to understand how user sentiment shapes platform specific themes and to identify key factors driving positive, neutral, and negative opinions across both platforms. The dataset we used was collected from publicly available app reviews, and our team collaboratively designed the full workflow.

Each team member took responsibility for different components of the pipeline, including data cleaning, manual labeling, sentiment classification using both traditional ML and transformer models, and topic modeling using classical and modern algorithms. While the work was collaborative and interconnected, I led several major parts of the project related to both sentiment analysis and topic modeling.

Specifically, my individual contributions include:

- Conducting manual sentiment labeling for the Threads dataset and performing model comparison based on these labels.
- Developing the Logistic Regression based sentiment classification model, including feature extraction, training, and evaluation.
- Implementing and analyzing the LDA topic-modeling pipeline, including preprocessing design, coherence evaluation, visualization, and topic interpretation.

These components formed the core of my technical involvement in the project, and they directly informed the overall interpretation and final insights presented in our group report.

Description of My Individual Work

1. Manual Labeling and Evaluation of Self-Supervised Sentiment Labels

An important part of my individual contribution was manually labeling approximately 3,000 Threads reviews to create a reliable benchmark for evaluating VADER's pseudo-labels. Since our sentiment-aware topic modeling pipeline depends heavily on accurate sentiment assignments, I carefully annotated each review as positive, negative, or neutral using a consistent rubric to reduce subjectivity. In parallel, I worked with Rachana to improve VADER's labeling accuracy by developing a hybrid thresholding approach. We explored methods for moving beyond fixed VADER score cutoffs and identified an open-source grid-search method that informed our design of a VADER + KNN hybrid strategy. My role involved interpreting VADER outputs against my human-labeled dataset and integrating the optimized thresholds to refine the pseudo-labels.

Using the manually annotated subset, I evaluated VADER's performance and found moderate alignment with human judgment, with an overall accuracy of 0.54 and a weighted F1-score of 0.52. While VADER excelled at identifying positive sentiment, its weaker recall for neutral and negative classes revealed a strong positivity bias and difficulty handling nuanced

expressions. These findings were crucial for shaping the rest of our workflow, demonstrating the limitations of lexicon-based methods and motivating the need for more robust machine learning models. Ultimately, my work in manual labeling, evaluation, and hybrid thresholding provided the foundational ground truth that guided model selection and informed downstream sentiment analysis and topic modeling decisions.

2. Sentiment Classification Using TF-IDF + Logistic Regression

Another major component of my individual contribution was the development of the TF–IDF + Logistic Regression sentiment classification model**.** This model served as our classical machine-learning baseline for evaluating how well a simple linear classifier performs when trained on VADER-generated pseudo-labels. My work involved designing the end-to-end pipeline, starting from feature extraction and extending through model training, hyperparameter selection, and performance evaluation on both the Threads and Twitter datasets.

To prepare the text data for modeling, I implemented TF–IDF vectorization to convert reviews into sparse numerical representations that capture word importance across the corpus. I then trained a multinomial Logistic Regression classifier on these features using the VADER pseudo-labels as supervision. The goal was to establish a performance benchmark that would help us understand whether a lightweight, interpretable model could effectively capture sentiment patterns in review text, and to determine how much additional benefit more advanced transformer-based models might provide.

Experiment. The workflow consisted of converting text reviews into TF–IDF feature vectors and training a multinomial Logistic Regression classifier using VADER-generated pseudo-labels. Hyperparameters such as regularization strength, solver choice, and n-gram range were tuned using cross-validation to ensure stable performance across the dataset.

Platform	**Accuracy**	**Weighted F1 Score**
Threads	0.74	0.69
Twitter	0.84	0.83

Table 1.0: Logistic Regression Comparision on Both Platforms

As seen in Table 1.0, the Logistic Regression model demonstrated strong performance given its simplicity. On the Threads dataset, it achieved an accuracy of 0.74 and a weighted F1-score of 0.69, significantly outperforming VADER's alignment with human labels. On the Twitter dataset, performance improved further, reaching 0.84 accuracy and 0.83 weighted F1. These results show that a linear TF–IDF-based classifier can learn meaningful sentiment distinctions from noisy pseudo-labeled data particularly on Twitter, where sentiment expression is more direct and polarized.

3. Topic Modeling Using Latent Dirichlet Allocation (LDA)

A major portion of my individual contribution involved developing and analyzing the LDA topic modeling pipeline for both Threads and Twitter. I implemented LDA as our baseline approach to examine how sentiment-specific themes differ across platforms. Since our project aimed to perform sentiment-aware topic modeling, I applied LDA separately to each sentiment subset positive, neutral, and negative for both platforms. This required designing a dedicated preprocessing workflow optimized for topic extraction rather than sentiment classification. I created a Gensim dictionary, filtered rare and overly common tokens using frequency thresholds (no_below=10, no_above=0.5), and converted all reviews into bag-of-words representations suitable for LDA training. I have clearly mentioned the algorithm (Algorithm 1) for the LDA, I used for our Topic Modelling Approach.

To ensure robustness, I trained multiple LDA models for each sentiment category, experimenting with different topic numbers ($k \in [3,6]$) and using alpha = "auto" with several passes for improved stability.

Algorithm 1: LDA Topic Modelling

INPUT: Preprocessed reviews for sentiment s => {pos, neu, neg}
OUTPUT: Set of k topics for each sentiment subset

1: Build Gensim dictionary from tokenized reviews.
2: Filter tokens using frequency thresholds:
    no_below = 10, no_above = 0.5
3: Convert corpus to bag-of-words representation.
4: For each sentiment subset:
    For k in {3, 4, 5, 6}:
    Train LDA model with:
    - Alpha = 'auto'
    - Multiple passes for stability
    Extract Top words per topic
5: Select the most coherent model for interpretation

For **Threads**, (From Table 2.0) positive reviews centered on general satisfaction, comparisons favoring Threads over competing platforms, and polite feature requests reflecting early-stage optimism. Neutral topics were dominated by functional observations, platform comparisons, and discussions about Meta's strategy, including references to Twitter and Elon Musk. Negative topics, however, were strongly characterized by crashes, reliability failures, missing features, and broader rejection of the product indicating that user dissatisfaction on Threads is primarily tied to functionality and technical stability.

Positive (coherence score ~ 0.56)	Neutral (coherence score ~ 0.58)	Negative (coherence score ~ 0.52)
General positivity and praise	*Platform comparisons and copying discourse*	*Crashes and reliability failures*
Comparisons favoring Threads	*Functional/technical notes*	*Low quality experience and missing features*
Polite feature requests	*Twitter vs Elon Musk narrative*	*Strong rejection of the product*
Platform approval		

Table 2.0: Topics Discovered by LDA for each Sentiment Label for Threads

In contrast, **Twitter** (From table 3.0) exhibited more emotionally intense and identity-driven discussions. Positive topics highlighted enthusiasm for features, perceived improvements under new ownership, and satisfaction with functionality. Neutral topics were more varied, covering leadership commentary, platform behavior, rebranding, feature removals, and ongoing usability changes. Twitter's negative topics were the most polarized, emphasizing disapproval of the "X" rebranding, leadership criticism, update frustrations, and dissatisfaction with recent platform decisions. These topic distributions demonstrate that while Threads sentiment revolves around usability and feature completeness, Twitter sentiment is shaped by broader platform identity and policy changes.

Positive (Coherence score ~ 0.54)	Neutral (Coherence score ~ 0.57)	Negative (Coherence score ~ 0.40)
General enthusiasm & enjoyment	*Leadership & feature commentary*	*Usability & update complaints*
Positive comparisons & improvements	*Platform behaviour and quality*	*Strong disapproval of rebranding*
Satisfaction with functionality or changes	*Functional notes on usage*	*Leadership-focused criticism*
	Rebranding & logo discussion	
	Updates & usability	
	Logo/name change mentions	

Table 3.0: Topics Discovered by LDA for each Sentiment Label for Twitter

Results

To gain an initial qualitative understanding of the language patterns associated with each sentiment category, I generated word clouds separately for positive, negative, and neutral reviews for both Threads and Twitter. These visualizations highlight the most frequent terms used by users and reveal platform-specific differences in how sentiment is expressed.

As we can see in Figure 1.0, In **Threads**, negative reviews prominently feature terms such as *"crash," "worst," "useless,"* and *"need,"* indicating that user dissatisfaction is largely driven by functional issues app instability, missing features, and poor usability. In contrast, positive reviews emphasize words like *"good," "great," "nice,"* and *"better,"* which suggests that praise is more general and less tied to specific features. Neutral comments frequently mention other platforms like *Instagram* and *Facebook,* as well as terms related to interface navigation such as *"feed," "follow,"* and *"option,"* reflecting discussions about app similarity, layout, and everyday usage rather than emotional expression.

Figure 1.0: Word Cloud For sentiment labels in Threads

Looking at Figure 2.0, For **Twitter**, the word clouds show a very different pattern. Negative reviews are dominated by *"bad," "terrible," "worst,"* and frequent mentions of *"Elon"* and *"Musk,"* indicating that user dissatisfaction is strongly tied to platform changes, policy decisions, and leadership rather than technical problems. Positive reviews, on the other hand, include terms such as *"good," "best," "great,"* and *"love,"* reflecting appreciation for platform features or updates. Neutral reviews again show heavy references to *"Elon Musk," "change," "new,"* and *"update,"* suggesting that even neutral discussions on Twitter are centered around ongoing platform transitions and rebranding rather than product functionality.

Figure 2.0: Word Cloud For sentiment labels in Twitter

Overall, the word cloud analysis highlights clear thematic differences between the two platforms: Threads sentiment is driven primarily by app performance and usability**,** whereas Twitter sentiment especially negative sentiment is shaped by leadership decisions, policy changes, and platform identity shifts**.** These early qualitative observations align closely with the subsequent quantitative sentiment classification and topic modeling results presented later in this report.

To obtain a more structured understanding of what users are talking about, we next apply **topic modeling**. Using LDA, we analyze the underlying themes within positive, neutral, and negative reviews for each platform. This allows us to move beyond surface-level word frequency and uncover the latent topics that drive user sentiment on Threads and Twitter.

On **Threads**, LDA surfaced clear and sentiment-specific topic structures. Positive reviews centered around general satisfaction, direct comparisons with Twitter, enthusiasm for the platform, and polite feature requests reflecting a user base excited about Threads but still hoping for improvements. Neutral reviews were dominated by observational commentary, especially comparisons to Twitter and Instagram, discussions of Meta's cloning strategy, and notes about bugs and missing features. Negative reviews on Threads focused heavily on technical failures such as crashes and posting errors, dissatisfaction with missing or poorly designed features, and broader rejection of the platform as unnecessary or low quality.

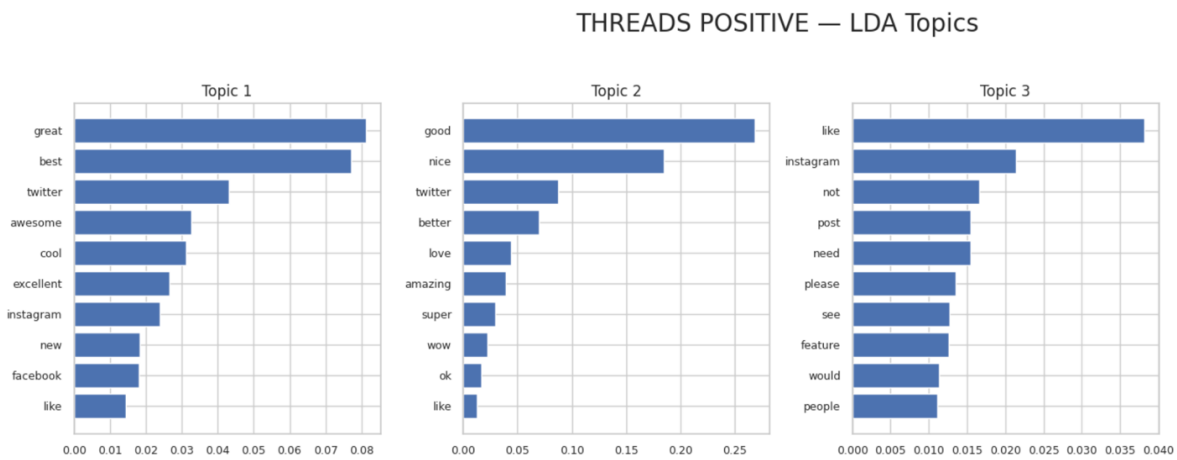

Figure 3.0: Top words for Positive Topics in Threads

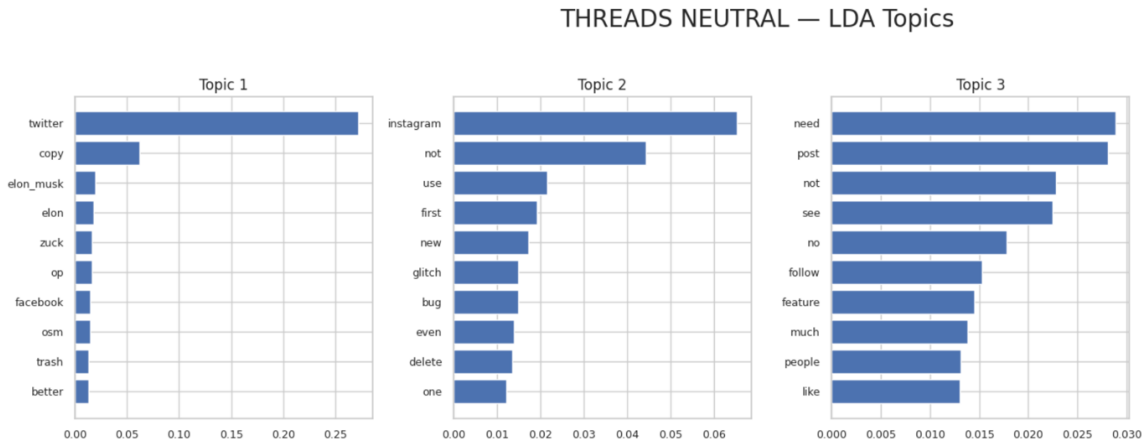

Figure 4.0: Top words for Neutral Topics in Threads

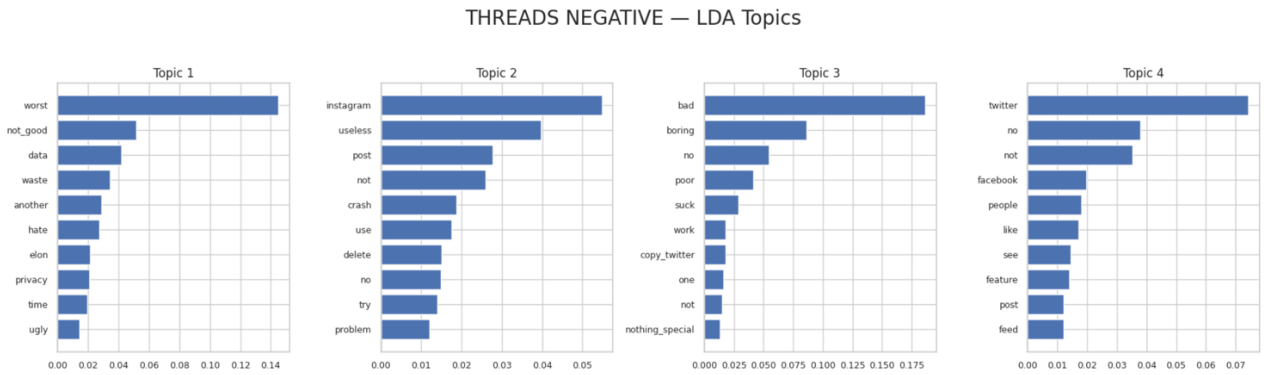

Figure 5.0: Top words for Negative Topics in Threads

On **Twitter**, the thematic patterns differed sharply, revealing stronger emotional intensity and a focus on identity and leadership. Positive reviews highlighted enjoyment of the platform, improvements under new ownership, perceived gains such as free speech, and satisfaction with functional updates. Neutral reviews were more fragmented and descriptive, covering topics such as rebranding, feature removals, usability changes, rate limits, and general observations about platform behavior. Negative reviews were particularly intense, centering on disapproval of the "X" rebranding, logo changes, rate-limit frustrations, and highly critical commentary directed at leadership.

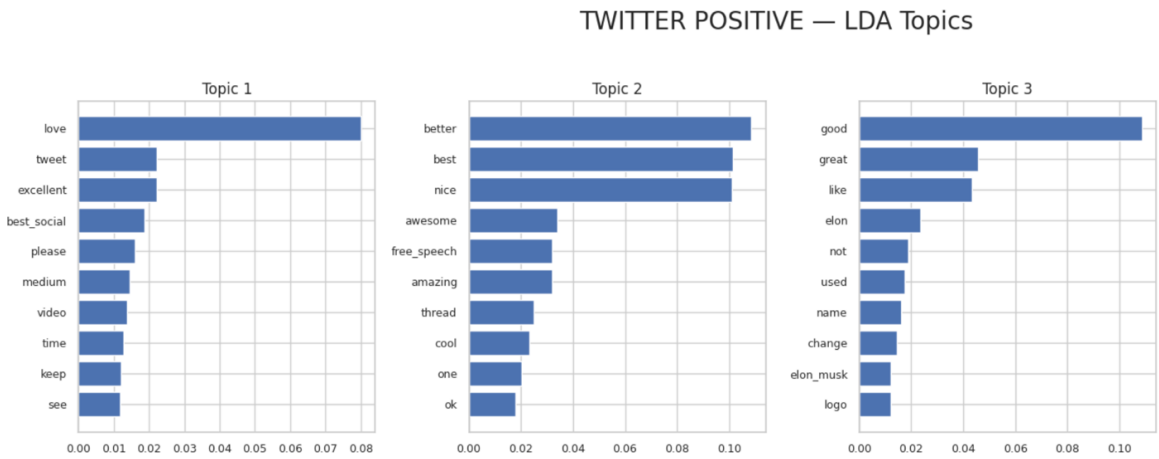

Figure 6.0: Top words for Positive Topics in Twitter

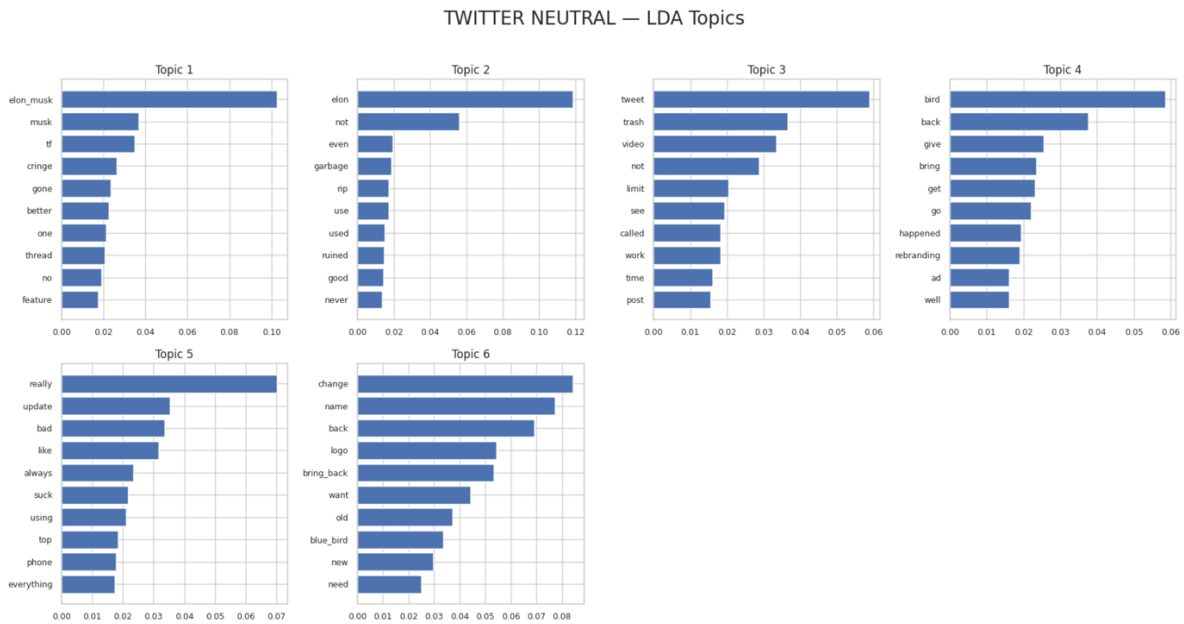

Figure 7.0: Top words for Neutral Topics in Twitter

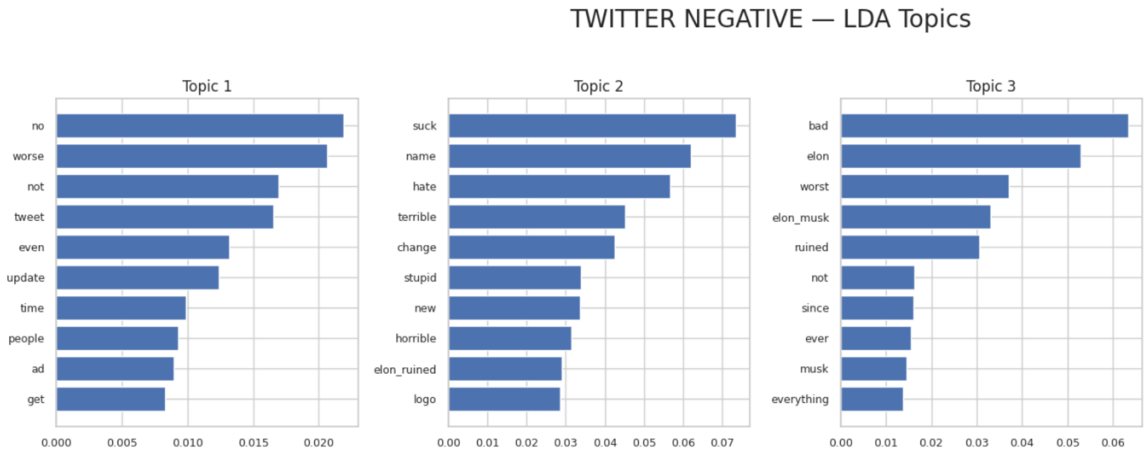

Figure 8.0: Top words for Negative Topics in Twitter

Overall, my LDA analysis served as the foundation for understanding how sentiment manifests in platform-specific discussions. It revealed that Threads negativity is driven by functional issues such as bugs and crashes, whereas Twitter negativity is driven by identity changes, leadership decisions, and policy shifts. These findings directly motivated the progression to more advanced topic modeling methods NMF and BERTopic (implemented by my teammates) which offer improved coherence and semantic richness. My work with LDA thus played a critical role in establishing benchmarks and shaping the direction of the remainder of the project's analysis.

Summary and Conclusion

In this project, I contributed to the sentiment and topic modeling components of a multi-stage NLP pipeline designed to analyze user reviews from Threads and Twitter. My work involved three major areas:

- manual sentiment annotation and evaluation of VADER's pseudo-labels
- development of a classical machine-learning sentiment classifier using TF–IDF and Logistic Regression
- implementation and interpretation of sentiment-aware LDA topic modeling for both platforms.

From the manual labeling experiment, I learned that lexicon-based methods like VADER struggle with informal, context-dependent social-media language. VADER achieved only moderate alignment with human labels (accuracy 0.54; F1 0.52), particularly failing to distinguish neutral and negative sentiment. This demonstrated the importance of human-curated ground truth and motivated the progression toward more robust learning-based models.

The TF–IDF + Logistic Regression classifier showed that even simple linear models can effectively learn sentiment boundaries from noisy pseudo-labels. The model significantly outperformed VADER, especially on Twitter (accuracy 0.84; F1 0.83), validating the usefulness of lightweight baselines and illustrating how textual features alone can capture sentiment differences across platforms.

The LDA analysis provided deeper insight into how sentiment manifests in user discussions. Threads topics were heavily centered on usability, reliability, and feature completeness, whereas Twitter topics reflected identity, leadership, and platform-wide policy changes. This contrast highlights how platform design and public discourse shape user expressions of sentiment. The LDA results also established essential baselines that guided our transition to more advanced topic modeling methods such as NMF and BERTopic.

Overall, this project demonstrated the importance of combining human supervision, classical machine learning, and topic modeling to achieve a rich understanding of platform-specific user sentiment. Through this work, I learned how model performance depends not only on algorithmic sophistication but also on data quality, labeling strategy, and domain context.

Code found on internet: Approximately 40 – 45 % of code was found on the internet, considering modification and addition of my own code.

References

[1] VADER optimal thresholds. (2020).Kaggle. https://www.kaggle.com/code/pawanbhandarkar/training-a-sith-lord

[2] A. Author(s), *"Comprehensive Evaluation of LDA, NMF, and BERTopic's Performance on News Headline Topic Modeling,"* **Journal of Computing Theories and Applications**, vol. 2, no. 2, pp. 268–289, Nov. 2024, doi: 10.62411/jcta.11635.

[3] Interpreting Word Clouds in Social Media Text (2024). https://www.crawlspider.com/how-to-use-word-clouds-in-social-media-strategy/

[4] A. K. Mishra, R. K. Barik and R. Sharma, "Sentiment Analysis Using VADER and BERT: A Comparative Study," *2020 International Conference on Data Science and Engineering (ICDSE)*, Patna, India, 2020, pp. 1–6, doi: 10.1109/ICDSE50453.2020.00009.